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Effects of spherical aberrations on micro welding of glass using ultra short laser pulses

Kristian Cvecek^{a,b,*}, Isamu Miyamoto^b, Matthias Adam^a and Michael Schmidt^{a,b,c}

^a*Bayerisches Laserzentrum GmbH, Konrad-Zuse-Straße 2-6, 91052 Erlangen, Germany*

^b*Erlangen Graduate School in Advanced Optical Technologies (SAOT), Friedrich-Alexander-Universität Erlangen-Nürnberg, Paul-Gordan-Straße 6, 91052 Erlangen, Germany*

^c*Chair of Photonic Technologies, Friedrich-Alexander-Universität Erlangen-Nürnberg, Paul-Gordan-Straße 3, 91052 Erlangen, Germany*

Abstract

In micro-welding of glass by means of ultra-short laser pulses focusing objective lenses with a high numerical aperture are necessary to generate plasma by nonlinear processes. However, due to the strong focus through the plane surface of the glass samples strong spherical aberrations occur, which increase the necessary power to start the nonlinear absorption processes. The influence of the spherical aberrations on the shape of the molten zone, on the pulse energy necessary for plasma generation is studied experimentally in the present work.

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1. Introduction

With the increasing availability of ultra short pulsed lasers processing of transparent materials has become more popular since with this laser type nonlinear effects can be utilized that allow for a well defined and precise in-depth processing of transparent dielectrics. E.g., ultra fast lasers can be used for glass welding by exploiting the fact that a microscopically small molten zone can be generated which does not exhibit overlarge mechanical stresses making defect-free glass welding possible [1].

* Corresponding author. Tel.: +49-9131-97790-37 ; fax: +49-9131-97790-11 .
E-mail address: k.cvecek@blz.org .

However, even for glass welding by ultra fast lasers tight focusing is advisable in order to have good control of the nonlinear effects. This can be accomplished by using focusing objectives with a high numerical aperture (NA) on the order of 0.5 and larger. When the focus of such an objective (which is assumed to provide spherical wavefronts in air) is placed into an optically denser material through a plane surface then so called spherical aberrations occur, which increase – among other effects – the focal spot diameter [2].

The formation of spherical aberrations occurs because the often-used paraxial approximation is no longer valid, since abaxial beams still carry a substantial portion of the beam. Instead, the Snellius' law has to be applied to describe the beam propagation at the plane surface. Such an analysis (compare fig. 1) shows that the refraction angles of abaxial beams tend with increasing incidence angle asymptotically towards the angle for total internal reflection of the transparent medium. This causes the abaxial beams to have their intersection at a much deeper position inside the material than the paraxial beams, thus increasing the focal volume of the beam. Since in ultrafast materials processing nonlinearities are exploited – that is effects that are highly intensity-dependent – one has to be aware of the changes caused by the spherical aberrations. We show in this paper that spherical aberrations also affect ultrafast laser glass welding and show which effects have to be expected.

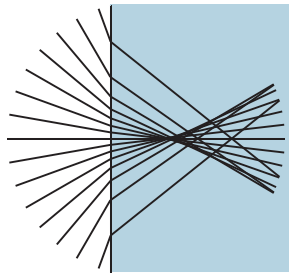


Fig. 1: Formation of spherical aberration when focusing a spherical wave front through a plane surface.

2. Experiment

The experimental setup used for the experiment is the same as used for glass welding [1] and is shown in fig. 2. The mode-locked laser (Fuego, Time-Bandwidth Company) has a FWHM-pulse duration of approx. 10 ps at 1064 nm. The repetition rate is variable and can be set to a maximum of 8.2 MHz. The laser can be focused onto the sample by microscope objectives with NAs of 0.4, 0.55, 0.8 and 0.9. The objectives are mounted on a vertical translation stage which allows to set the focus height of the objectives. The samples are ca. 1 mm thick D263 borosilicate glass plates positioned on an x-y translation stage.

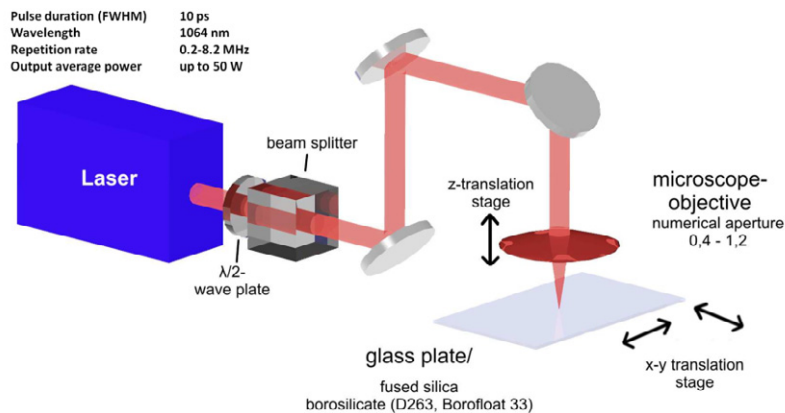


Fig. 2: The schematic of the experimental setup.

The first part of the experiment consisted of determining the necessary laser pulse energy for plasma generation at different depths inside the glass. The processing parameters were chosen to be 1 MHz pulse repetition rate and 200 mm/s feed speed. The high feed speed helps to suppress heat accumulation effects known to occur during glass welding [3]. For each experiment run with a different objective the objective was set with the vertical stage to have its focus on the upper surface of the sample. The objective was then displaced in steps by an amount of approx. 50 μm deeper into the glass. At each focusing depth the sample was set in continuous motion with the above feed speed while the average laser power was manually increased (with a care to achieve a smooth increase) until the laser power was sufficient to set off plasma generation. Since the nonlinear effects that occur during plasma formation generate isotropic white light the onset of plasma generation can be easily observed by human eye from the side even while wearing laser safety goggles. As soon as the plasma was registered the laser power increasing was ceased and the laser power was measured. For the objectives of NA 0.4 up to 0.8 the power measurement was performed after the objective (AO) providing the power by which the sample was irradiated. However, since the aperture angle of the 0.9 NA objective was too large to be measured by the power meter the laser power was measured in front of the objective (IFO). The losses in this objective are estimated to be roughly 90 % since it was not AR coated for 1064 nm, but due to its large wavelength design range (ultraplan) we assumed it to provide still an acceptable focusing. Moreover, this was the only objective with a correction ring for spherical aberrations so we had the opportunity to measure the effects of the spherical aberration compensation (SAC). For this task the correction ring was set to 0.11 and 0.23 corresponding to a setting generally used to compensate the spherical aberration of a coverslip glass used in microscopy with thicknesses of 110 μm and 230 μm . This has the disadvantage that at shallow depths <110 μm the objective is overcompensating. A further drawback of this objective is its short working distance, limiting measurements to a depth of ca. 300 μm .

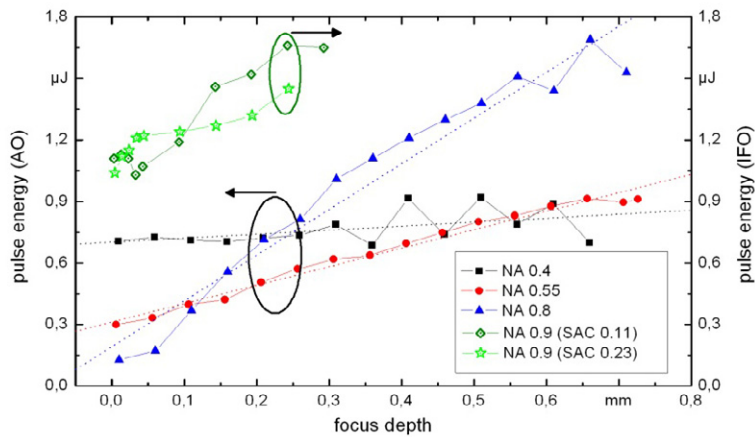


Fig. 3: Pulse energy necessary for plasma generation depending on focus depth. (Feed speed 200 mm/s, pulse repetition rate 1 MHz, AO – pulse energy measured after objective, IFO – pulse energy measured in front of objective, SAC – spherical aberration compensation).

The second part of the experiment consisted of providing melt runs in the material with the different objectives with the aim to evaluate the cross-section of the molten zone at different depths. The processing parameters were the same as before except for the average laser power. Here, the power was set by a factor of 3 respectively 2 larger than the measured plasma ignition power at the corresponding height for objectives with NA 0.4 respectively NA 0.55 – 0.9. The reason for this was to achieve slightly larger cross sections than the ignition power would provide in order make the evaluation of the melt runs with optical microscope easier and, more importantly, to provide approximately the same power density for the melt runs independently on their depth. After processing the samples were cut and polished to observe the cross section of the welding seam. The cross sections were evaluated by measuring the depth-dependent aspect ratio (height/width).

3. Results and Discussion

Fig. 3 shows the laser pulse energy necessary to start plasma at different depths. For all NAs the necessary pulse energy increases with increasing depth. The pulse energy increase can be approximated by a line equation in good agreement with the measured values. As expected, the slope increases strongly with the NA, since the spherical aberrations increase for high NAs. Of course, due to the tighter focusing of high NA objectives the power necessary to start plasma in shallow depths is smaller for high NAs. This leads to the interesting fact that due to the higher slope high NA objectives need a much higher power to set plasma off than low NA objectives at a certain depth, which can be for some cases quite contra productive.

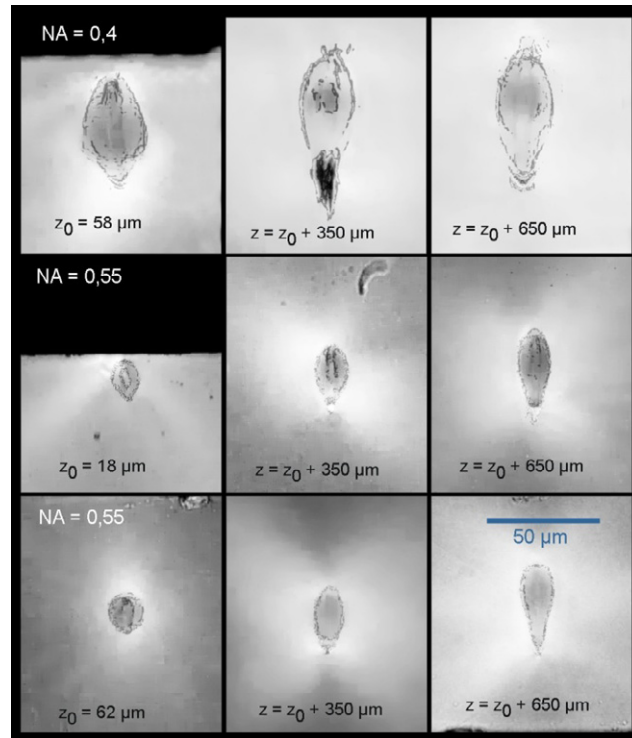


Fig. 4: Cross sections of the welding seams depending on focus depth and used NA.

However, the increase of the power necessary for plasma generation is not the only effect of spherical aberrations. Additionally, the shape and volume of the molten zone changes which is shown by cross sections in fig. 4. Here, the images of the cross sections were post-processed in order to enhance the visibility of their boundaries as it is generally difficult (due to small differences in the refractive indices between molten zone and rest of the material) to obtain clear pictures at high magnification factors in a conventional optical microscope. Keeping in mind that the power density in the focal volume was kept approximately constant independently of the focus depth, one can see that the molten zone elongates and grows larger for deeper focusing depth. Especially at higher numerical apertures a small tip appears at the bottom of the molten zone. This might prove potentially problematic for glass welding due to the high cooling rates of the molten zone at the tip since the tip has a larger boundary area than a spherically shaped molten zone consequently leading to a higher heat flux. In turn the resulting lower temperatures of the tip may provide less stable welding seams or even cracks due to the brittleness of the cold glass material [4].

In order to quantitatively analyze the shape of the molten zone the aspect ratio of the molten zones (height/width) was measured. The results are shown in fig. 5. It can be seen that the aspect ratio differs at most for shallow depths between the NAs of the objectives. For deeper focusing there is almost no nominal difference between the NAs. As the results of the 0.9 NA objective with SAC show, SAC helps to compensate spherical aberration and provide more symmetrical molten zone shape even in a larger depth.

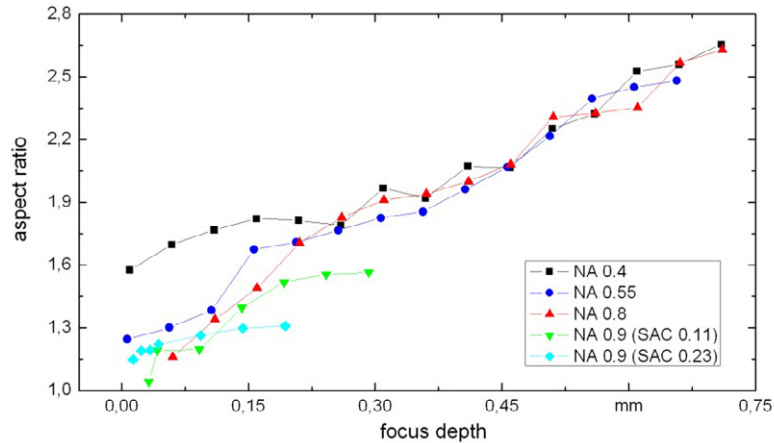


Fig. 5: Aspect ratio of the cross sections of the welding seams depending on focus depth and used NA (Feed speed 200 mm/s, pulse repetition rate 1 MHz, SAC – spherical aberration compensation).

The presented results show that it is important to take spherical aberrations into account when welding glass by ultra short pulsed lasers especially if the welding has to be done at a larger depth. In this case higher pulse energies and, consequently, higher average powers are necessary to start plasma. Furthermore spherical aberrations increase also the aspect ratio of the molten zone and lead to the formation of a tip at the bottom of the molten zone. This asymmetry is expected to lead to a degradation of the welding seam quality since the asymmetry causes the molten zone to have a larger surface area resulting in higher cooling rates. Both, the intensity reduction and higher cooling rates due to spherical aberration can be compensated by higher pulse energies. But, this might result in turn in reduction of the parameter range where a stable glass welding process is possible. One possible option the experimental results hint at is to reduce the numerical aperture. However, future experiments must prove if the welding seam quality, e.g. frequency and type of defects, can be kept on the same level as when high NA objectives are used in a more shallow depth.

Acknowledgements

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